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ORIGINAL ARTICLE

ACTA2 Pathogenic Variants Activating Heat Shock Factor 1 and Increasing Cholesterol Biosynthesis in Smooth Muscle Cells Predispose to Early Onset Atherosclerosis

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BACKGROUND: *ACTA2* pathogenic variants predispose to thoracic aortic disease, and a subset of variants lead to early onset atherosclerotic cardiovascular disease (ASCVD). The molecular pathway linking misfolded SMA (α-smooth muscle actin) monomers to augmented atherosclerosis–associated smooth muscle cell phenotypic modulation can be modeled in vitro by stably expressing the *ACTA2* p.R149C variant in *Acta2*$^{-/-}$ smooth muscle cells.

METHODS: The Montalcino Aortic Consortium patient registry was used to identify cases with *ACTA2* pathogenic/likely pathogenic missense variants. These patients were surveyed, and their medical records were reviewed to identify cases with early onset ASCVD. The variants for these cases, as well as other recurrent *ACTA2* missense variants, were individually expressed in *Acta2*$^{-/-}$ smooth muscle cells, and transcript and protein levels, HSF1 (heat shock factor 1) activation, HMGCR (3-hydroxy-3-methylglutaryl-coenzyme A reductase) expression and activity, cholesteryl ester levels, and downstream smooth muscle cell phenotypic modulation were assessed.

RESULTS: Early onset ASCVD included coronary artery disease, peripheral vascular disease, and atherosclerotic plaques identified by imaging in the arch, descending, or abdominal aorta, along with the celiac, iliac, renal, or vertebral arteries. Twelve *ACTA2* variants were identified to be associated with early onset ASCVD. Early onset ASCVD was correlated with HSF1 activation (P=0.035), cellular cholesteryl ester levels (P=0.0031), and having one family member with the specific *ACTA2* pathogenic variant who had early onset ASCVD (P=0.0001).

CONCLUSIONS: Assays assessing the molecular mechanism that leads to early onset ASCVD can identify which *ACTA2* pathogenic variants will trigger this condition. Ultimately, this information informs precision medical care for individuals with *ACTA2* pathogenic variants, with the ultimate goal of preventing thoracic aortic disease and ASCVD.

Key Words: actins ■ atherosclerosis ■ cholesterol esters ■ coronary artery disease ■ heat shock transcription factors

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This manuscript was sent to Joshua W. Knowles, Guest Editor, for review by expert referees, editorial decision, and final disposition.

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Supplemental Material is available at https://www.ahajournals.org/doi/suppl/10.1161/CIRCGEN.125.005169.
For Sources of Funding and Disclosures, see page XXX.

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Nonstandard abbreviations and acronyms

ASCVD	atherosclerotic cardiovascular disease
CAD	coronary artery disease
CCT	chaperonin-containing T-polypeptide complex
CE	cholesteryl ester
HMGCR	3-hydroxy-3-methylglutaryl-coenzyme A reductase
HSF1	heat shock factor 1
oxLDL	oxidized low-density lipoprotein
PERK	protein kinase R-like ER kinase
PV	pathogenic/likely pathogenic variant
SMA	α-smooth muscle actin
SMC	smooth muscle cell
WT	wild type

ACTA2 encodes the smooth muscle cell (SMC)–specific isoform of α-actin (SMA [α-smooth muscle actin]), which polymerizes to form the thin filament of SMC contractile units. Heterozygous missense variants in *ACTA2* confer a highly penetrant risk for thoracic aortic aneurysms and dissections, termed heritable thoracic aortic disease.[1] *ACTA2* missense variants disrupt the dynamics of SMA filament formation and its interaction with myosin filaments, resulting in decreased force generation by SMCs.[2,3] Pathogenic/likely pathogenic variants (PVs) in other genes that predispose to heritable thoracic aortic disease similarly disrupt SMC force generation, suggesting a common molecular mechanism.[4,5] Given that SMA is a cytosolic contractile protein and *ACTA2* expression is limited to SMCs and myofibroblasts, it was an unexpected discovery that a subset of *ACTA2* missense PVs, including *ACTA2* p.R149C, the most common PV in the gene, also predisposes to early onset coronary artery disease (CAD) in the absence of atherosclerotic risk factors (defined CAD ≤55 years in males and 65 years in females).[6,7]

Single-cell transcriptomics has identified complex phenotypic modulation of SMCs associated with plaque formation in hypercholesterolemic mice characterized by decreased expression of differentiation markers and variably increased expression of markers for stem cells, fibroblasts, macrophages, and osteogenic-like cells.[8–10] SMCs exposed to free cholesterol or oxLDL (oxidized low-density lipoprotein) in vitro recapitulate much of this phenotypic modulation, and we initially determined that cholesterol enters the endoplasmic reticulum and triggers endoplasmic reticulum stress, with modulation driven by downstream PERK (protein kinase R-like ER kinase) signaling.[11,12] SMC-specific deletion of *Perk* in hyperlipidemic male mice (*Perk*$^{SMC-/-}$) reduces atherosclerotic plaque formation by up to 80% compared with wild-type (WT) mice, emphasizing the importance of SMC PERK signaling in plaque formation.[13] We identified that the *ACTA2* p.R149C variant also increased SMC intracellular cholesterol levels and PERK signaling through a unique molecular mechanism and augmented plaque burden in an *Acta2*$^{R149C/+}$ mouse model.[14,15] Briefly, polypeptides of SMA monomers are synthesized by ribosomes and then directly shuttled into the CCT (chaperonin-containing T-polypeptide complex) complex, which is a large, multimeric complex that folds all actins.[16] Monomers of SMA with the p.R149C variant are misfolded and retained in the CCT complex in SMCs, and this retention activates cytosolic stress and HSF1 (heat shock factor 1) in *Acta2*$^{R149C/+}$ SMCs.[14,15] HSF1 signaling augments cholesterol biosynthesis in the mutant SMCs, in part through increased level and activity of a rate-limiting enzyme for cholesterol biosynthesis, HMGCR (3-hydroxy-3-methylglutaryl-coenzyme A reductase). As intracellular levels of cholesteryl esters (CEs) increase, endoplasmic reticulum stress and PERK are increased, driving augmented ATF4 levels and KLF4 activity.[8,12,13,17] Specifically, *Acta2*$^{R149C/+}$ SMCs dedifferentiate, increase proliferation and migration, and upregulate the expression of markers of other cell types, including macrophage and chondrocyte, whereas WT SMCs do not.[12,14,17] *Acta2*$^{R149C/+}$ mice crossed into an *Apoe*$^{-/-}$ background have a 2.5-fold increased burden of atherosclerotic plaque compared with *Apoe*$^{-/-}$ mice when fed a high-fat diet without differences in plasma lipid levels, thus replicating the increased burden observed in patients with the *ACTA2* p.R149C variant. Because inhibition of HMGCR reduces cholesterol biosynthesis, *Acta2*$^{R149C/+}$*Apoe*$^{-/-}$ mice were treated with pravastatin, which reduced plaque burden to levels similar to the *Apoe*$^{-/-}$ mice, despite similar reductions in plasma lipid levels in both mice.[14] Premature onset of atherosclerotic cardiovascular disease (ASCVD) in individuals with *ACTA2* p.R149C occurs in the absence of risk factors, and the molecular pathway driving increased plaque burden can be blocked with statins even if lipid levels in the blood are normal. Thus, it is critical to identify all *ACTA2* missense PVs that activate this pathway to prevent ASCVD and its complications in individuals harboring these variants.

We demonstrated that the molecular pathway linking misfolded SMA monomers to augmented atherosclerosis–associated SMC phenotypic modulation can be recapitulated by stably expressing the *ACTA2* p.R149C variant in mouse SMCs that do not express *Acta2* (*Acta2*$^{-/-}$ SMCs).[14] Using this assay system, we assessed if *ACTA2* missense variants are associated with activation of HSF1, increased cholesterol biosynthesis, CE accumulation, and augmented SMC phenotypic modulation. Combining these data with vascular disease data from families with *ACTA2* PVs precisely defines which *ACTA2* PVs trigger ASCVD and which do not. Ultimately, these studies provide critical information as to which

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ACTA2 PV carriers need surveillance and treatment to prevent early onset ASCVD and its complications.

METHODS

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Survey Creation and Distribution

The Montalcino Aortic Consortium registry of patients with variants in known aortopathy genes was approved by the institutional review board at the University of Texas Health Science Center at Houston and at other Montalcino Aortic Consortium sites involved in this study. Informed consent was obtained from all study participants. From an initial cohort of 387 patients with *ACTA2* variants, 42 unique *ACTA2* variants were confirmed to be PV in 226 patients, using American College of Medical Genetics guidelines in May 2022 (Supplemental Material).[18] Of these 226 patients with *ACTA2* PVs, 57 patients (living and deceased) had the p.R149C variant, and medical records for 47 of them were reviewed for ASCVD burden and complications. These results were used to create an electronic survey to assess ASCVD burden in participants with other *ACTA2* missense PVs, using questions adapted from the San Diego Claudication Questionnaire for peripheral vascular disease and PhenX, a database of protocols optimized for data collection in clinical research (Supplemental Material).[19,20] Branching logic was utilized to minimize survey completion time. *ACTA2* PV patients aged ≥18 years were sent the survey via email, including carriers with the p.R149C variant for validation. Survey data were collected using REDCap electronic data capture tools.[21,22] When a participant reported ASCVD burden or complication in a survey response, medical records were reviewed for verification. For participants who reported diagnoses of atherosclerosis in multiple arteries (see section Other Atherosclerosis on the survey in the Supplemental Material), the age at which the first diagnosis was made was used as the age of onset. For the *ACTA2* PV patients who were unable to receive or complete the survey, atherosclerotic events and plaque burden were extracted from their medical histories provided at enrollment and medical records when available.

Atherosclerosis Inclusion Criteria

Early onset ASCVD was identified by CAD, myocardial infarction, peripheral vascular disease, or other arterial plaque, diagnosed <55 years in males and < 65 years in females, in accordance with the age thresholds for premature ASCVD used in the American College of Cardiology's Guideline on Blood Cholesterol Management.[23] In all reviews of medical records, explicit diagnoses of CAD, myocardial infarction/heart attack, and peripheral vascular disease were counted as these respective atherosclerotic events. Myocardial infarction, TIA, or stroke events secondary to aortic dissection or surgery were excluded as atherosclerotic events. Because a subset of *ACTA2* missense PVs predispose to moyamoya-like occlusion of the large arteries in the brain, TIA and stroke diagnoses were not included as ASCVD for this study.[24–26] Aortic events were defined as type A or B aortic dissections or repairs of thoracic aortic aneurysms.

Expression and Protein Levels Analyses of *ACTA2* Missense Variants in *Acta2*$^{-/-}$ SMCs

PVs seen with ≥1 case of early onset ASCVD were assayed in cellular studies, along with WT *ACTA2*, PVs in ≥5 patients, and PVs with no reported incidence of ASCVD as controls. *ACTA2* missense variants were individually introduced into the WT mouse *Acta2* gene by site-directed mutagenesis, and the *Acta2* cDNA construct was subcloned into the lentiviral vector pLVX-zsGreen1 (Clontech) and used to infect SMCs explanted from *Acta2*$^{-/-}$ mice, as described previously.[14] SMCs were explanted in accordance with protocols approved by the Animal Welfare Committee of the University of Texas Health Science Center at Houston. Total RNA isolation and qRT-PCR were performed as described earlier, with *Gapdh* as internal control (Applied Biosciences) when using Taqman chemistry (Quantabio) and with 18S rRNA as internal control (Millipore Sigma) when using SYBR green chemistry (Quantabio).[13,14] To determine the level of SMAs, infected cells were lysed in radioimmunoprecipitation buffer (Millipore Sigma), containing protease inhibitor cocktail (GenDEPOT), followed by centrifugation to obtain clear cell lysates. Protein concentration in the lysates was quantified using Bradford reagent (Bio-Rad Laboratories), and the lysates were subjected to sodium dodecyl sulfate polyacrylamide gel electrophoresis, transferred to polyvinylidene fluoride membranes, blocked with 5% BSA in tris-buffered saline containing 0.01% Tween 20, and then incubated overnight at 4 °C with α-SMA (Abcam) and GAPDH (Cell Signaling Technology) antibodies. The membranes were washed with 0.01% Tween 20, incubated with bovine anti-goat or goat anti-rabbit horseradish peroxidase-conjugated secondary antibodies (Jackson Immunoresearch), and then washed with 0.01% Tween 20. Blots were visualized using chemiluminescence (GE Healthcare). Details of the antibodies used are provided in Table S1.

Assays to Assess Misfolding of the SMA Monomers and Downstream Pathways Activated

Immortalized *Acta2*$^{-/-}$ mouse SMCs stably expressing WT or individual *Acta2* missense variants were transfected with HSF1 or KLF4 Cignal Reporter plasmid (Qiagen) and incubated for 48 hours. Luciferase activity, a measure of transcription factor binding to the DNA, was measured using a Dual Luciferase Assay kit (Promega), and transcriptional activity was expressed as a ratio of firefly to internal Renilla chemiluminescence, as described earlier.[13,14] HMGCR enzymatic activity was estimated in *Acta2*$^{-/-}$ SMCs expressing *Acta2* variants using an HMGCR activity assay kit, as described previously.[14] Immortalized *Acta2*$^{-/-}$ mouse SMCs stably expressing individual *Acta2* missense variants were incubated overnight at 37 °C. A colorimetric assay was used to estimate cellular CEs using a Cholesterol/Cholesteryl Ester Quantitation Assay kit (Colorimetric/Fluorometric, Abcam), according to the manufacturer's instructions.[14] Gene expression was quantified using SYBR Green or Taqman chemistry (SYBR Green primers are provided in Table S2).

Skin samples, either from punch biopsy or surgical incision sites, were stored in DMEM high-glucose media supplemented with 10% FBS and antibiotic/antimycotic (complete DMEM).

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Tissue was washed with 70% ethanol, then PBS, and then in complete DMEM; fat and epidermal layers were removed; and the remaining dermal tissue was minced into 1-mm pieces, then sandwiched between a tissue culture dish and a sterile glass coverslip, and fed with complete DMEM to promote outgrowth. Once dermal fibroblast cultures were established, they were cultured in a monolayer in complete DMEM for up to 6 passages. All the variant fibroblasts used in the in vitro studies were isolated from White individuals aged 20 to 74 years, including 2 females, while the 3 control fibroblasts were obtained from a White male, a White female, and a Hispanic female, aged 41 to 53 years.

Statistical Analyses

Time-to-clinical event curves were calculated according to the method of Kaplan and Meier using GraphPad Prism 10.6.0. For the cellular studies, data were tested for normality using GraphPad Prism 10.6.0. All data that passed normality were analyzed with 1-way ANOVA, followed by the Dunnett multiple comparisons test for multiple group comparisons and with the Student *t* test followed by the Welch correction when comparing 2 groups. When the data did not pass normality, they were analyzed with the Mann-Whitney *U* test when comparing 2 groups. To determine the association between ASCVD and signaling pathway components, protein expression (for SMA) and CEs for each variant were averaged across 3 experiments and compared with the absence (0) and presence (1) of ASCVD using the Student *t* tests followed by the Welch correction (when the data were distributed normally) or by the Mann-Whitney *U* test. For determining the association of ASCVD with phenotypic modulation, the expression of phenotypic modulation markers *Lgals3*, *Spp1*, and *Fn1* was each assigned a score of 1 if the mRNA fold change was significantly greater compared with WT *ACTA2* and 0 if not. The scores for each marker were then summed to calculate a total phenotypic modulation score for each variant, which was subsequently correlated with the presence of ASCVD using the Student *t* tests or the Mann-Whitney *U* test. For associations analyzed using the Spearman correlation, HSF1 activity was correlated with ASCVD and with family members' ASCVD status. A corrected $P<0.05$ was considered statistically significant.

RESULTS

Study Population and Extent of Early Onset ASCVD in *ACTA2* p.R149C Study Participants

The study population from the Montalcino Aortic Consortium comprised 226 participants (the 47 living/deceased p.R149C participants with medical records plus the 179 survey-eligible participants) with a mean age of 43.08 years (±18.59 years; Table S1). The distribution of sex assigned at birth was 43.4% female (n=98) and 56.6% male (n=128). Regarding race, the majority identified as White (80.1%, n=181), followed by smaller proportions identifying as Hawaiian Native or other (4.4%, n=10), Asian (3.5%, n=8), >1 race (1.3%, n=3), Black (0.4%, n=1), and American Indian or Alaska Native (0.4%, n=1), while, for 9.7% (n=22), race was not reported. Most participants were not Hispanic or Latino (88.1%, n=199), with 5.3% (n=12) identifying as Hispanic or Latino and 6.6% (n=15) not responding (Table S3).

Seventeen of the 47 living/deceased *ACTA2* p.R149C patients with medical records (36.2%) had been diagnosed with early onset ASCVD, and 23 of 47 patients had experienced an aortic event (48.9%), which were all presentation with a type A or B dissection.[27] Of the 47 patients, 9 patients had both early onset ASCVD and aortic dissection, 8 had only early onset ASCVD, 14 had only an aortic event, and 16 had neither ASCVD nor aortic event (Figure S1). Early onset ASCVD in the p.R149C patients included the following: 11 cases of CAD, 4 cases of peripheral vascular disease, and 7 cases of atherosclerotic plaques noted in the arch, descending, or abdominal aorta on imaging, as well as single cases of plaques in the celiac, iliac, renal, or vertebral arteries. Time-to-event curves showed no significant difference in the age of onset of ASCVD versus aortic dissection for *ACTA2* p.R149C cases, with onset of either vascular disease rare in childhood, but increased events after the age of 20 years (Figure 1A). Six of the 47 patients died before the age of 55 years: 4 of aortic dissections, 1 of CAD, and 1 of an unknown cause.

American Heart Association.

Identification of *ACTA2* Missense PVs Seen With Early Onset ASCVD

Of the 226 *ACTA2* study participants overall, surveys were sent to the 106 participants who were aged ≥18 years and had available contact information (including *ACTA2* p.R149C); of these, 63 (59%) completed the survey. For the other 163 of 226 participants, a medical history review was performed. In total, 34 participants (15%) with early onset ASCVD were identified (Figure 1B). For comparison, of the 139 individuals from the Montalcino Aortic Consortium with *SMAD3* PVs, only 0.7% have early onset ASCVD (Table S4). Early onset ASCVD was reported in 18 of the 63 survey respondents, and diagnoses were confirmed by reviewing the individuals' medical records (Supplemental Material). Of these 18 survey respondents, 9 had *ACTA2* p.R149C, 2 had p.R212Q, 3 had p.R258H, and there was 1 case each for p.G270E, p.I371N, p.R198C, and p.E59K. For the other 163 participants, medical histories and medical records identified another 16 cases of early onset ASCVD. Additional *ACTA2* variants seen with early onset ASCVD included 2 patients with p.R185Q and 1 patient each with p.R39H, 1 p.R118Q, p.Y135H, and p.P166T. Arteries with atherosclerosis in patients with *ACTA2* p.R149C were similar to arteries involved in patients with other *ACTA2* variants (Table). As an example of ASCVD, an angiogram of a 57-year-old female participant with *ACTA2* p.R258H is presented, highlighting significant CAD, including high-grade stenoses of the proximal left circumflex coronary artery and multifocal stenoses of

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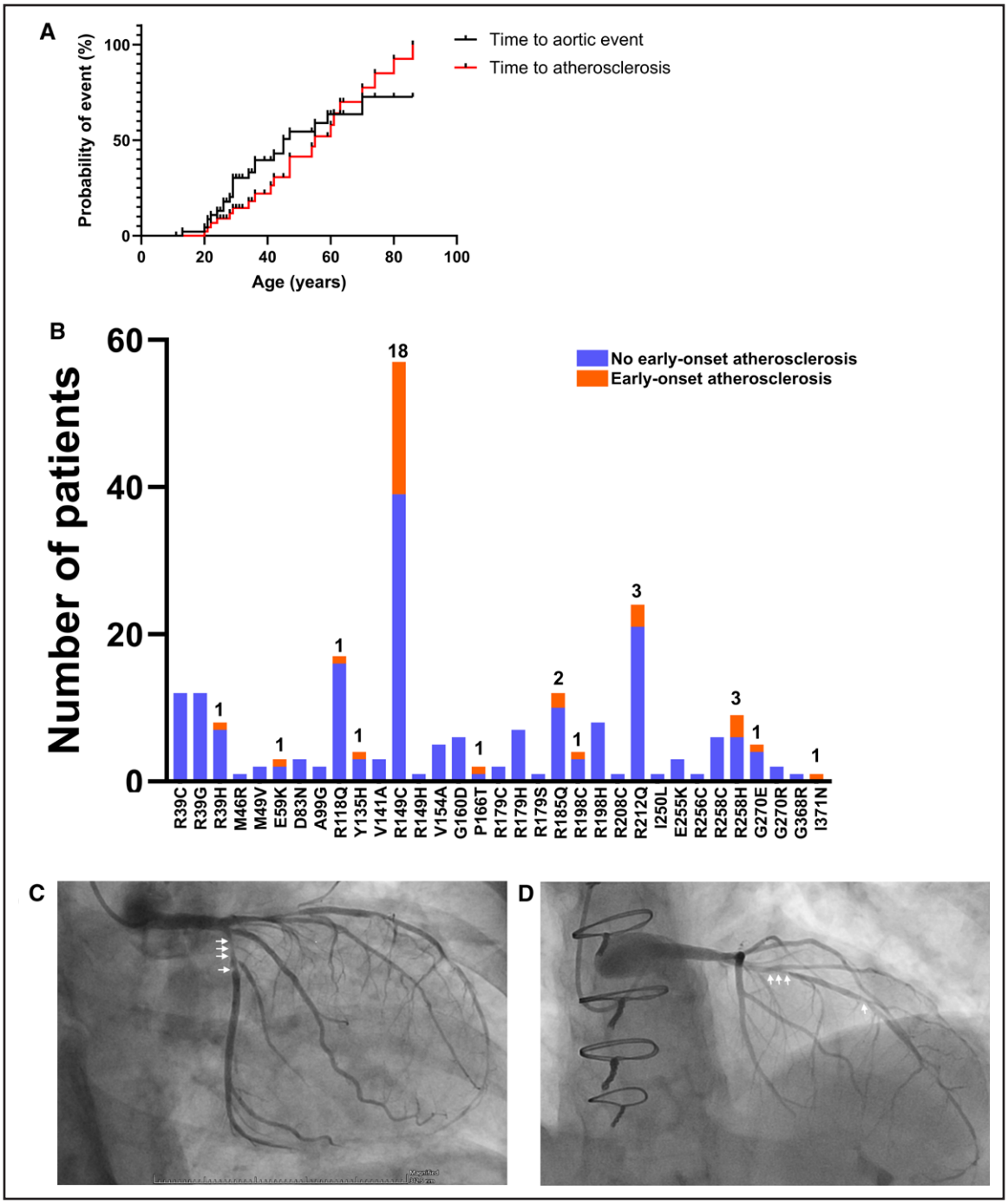

Figure 1. Atherosclerotic cardiovascular disease (ASCVD) and aortic events in *ACTA2* p.R149C patients, and a summary of *ACTA2* pathogenic/likely pathogenic variants (PVs) seen with early onset ASCVD.
A, The ages at which the 47 p.R149C patients had aortic dissections and were diagnosed with ASCVD were analyzed in Kaplan-Meier time-to-event curves. Each hatch mark represents the death of a patient or the censorship of a patient (the maximum age at which relevant information was known about the patient). **B**, Each of the 226 *ACTA2* patients was categorized as having or not having early onset ASCVD. The numbers above the bars state the number of patients with early onset ASCVD for each variant. **C**, A left coronary angiogram shows high-grade stenoses (arrows) of the proximal left circumflex coronary artery in a 57-year-old female patient with *ACTA2* p.R258H. **D**, A left coronary angiogram for the same patient shows multifocal stenoses (arrows) of the proximal and mid-left anterior descending coronary artery.

the proximal and mid-left anterior descending coronary artery (Figure 1C and 1D). Study participants with the following *ACTA2* variants had no evidence of early onset ASCVD: p.R39C (n=12), p.R39G (n=12), p.M46R (n=1), p.M49V (n=2), p.D83N (n=3), p.A99G (n=2), p.V141A (n=3), p.R149H (n=1), p.V154A (n=5), p.G160D (n=6), p.R179C (n=2), p.R179H (n=7), p.R179S (n=1), p.R198H (n=8), p.R208C (n=1), p.I250L (n=1), p.E255K

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Table. *ACTA2* Variants Seen With Early Onset ASCVD Have Increased HSF1 Activation and CE Levels

Variant	Total patients (percent with ASCVD)	Cases of CAD/MI	Cases of PVD	Cases of plaque in other arteries	HSF1 activation	CE levels
Variants seen with early onset ASCVD						
R149C	57 (32%)	12	5	15: aorta (7), carotid (3), iliac (2), celiac (1), renal (1), and vertebral (1)	Increased	Increased
R212Q	24 (13%)	1	0	3: aorta (1), carotid (1), and vertebral (1)	Increased	Increased
R118Q	17 (6%)	1	0	0	Decreased	Decreased
R185Q	12 (17%)	2	0	0	Increased	Increased
R258H	9 (33%)	1	1	5: aorta (3), arteries of the aorta (1), and iliac (1)	Increased	Increased
R39H	8 (13%)	1	0	0	Increased	Increased
G270E	5 (20%)	0	0	2: aorta (1) and carotid (1)	Increased	Increased
Y135H	4 (25%)	0	1	0	Increased	Increased
R198C	4 (25%)	0	1	0	Increased	Increased
E59K	3 (33%)	1	0	0	Increased	Increased
P166T	2 (50%)	1	0	0	Increased	Increased
I371N	1 (100%)	1	0	1:aorta	Increased	Increased
Variants assayed not seen with early onset ASCVD						
R39C	12 (0%)	0	0	0	No change	Decreased
R39G	12 (0%)	0	0	0	No change	No change
R198H	8 (0%)	0	0	0	No change	No change
G160D	6 (0%)	0	0	0	No change	Decreased
R258C	6 (0%)	0	0	0	No change	No change
V154A	5 (0%)	0	0	0	No change	No change
R149H	1 (0%)	0	0	0	Increased	Increased
Variants not assayed and not seen with early onset ASCVD						
R179H	7 (0%)	0	0	0		
D83N	3 (0%)	0	0	0		
V141A	3 (0%)	0	0	0		
E255K	3 (0%)	0	0	0		
M49V	2 (0%)	0	0	0		
A99G	2 (0%)	0	0	0		
R179C	2 (0%)	0	0	0		
G270R	2 (0%)	0	0	0		
M46R	1 (0%)	0	0	0		
R179S	1 (0%)	0	0	0		
R208C	1 (0%)	0	0	0		
I250L	1 (0%)	0	0	0		
R256C	1 (0%)	0	0	0		
G368R	1 (0%)	0	0	0		

Some patients had atherosclerosis involving multiple arteries, which were all included in the table. ASCVD indicates atherosclerotic cardiovascular disease; CAD, coronary artery disease; CE, cholesteryl ester; HSF1, heat shock factor 1; MI, myocardial infarction; and PVD, peripheral vascular disease.

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(n=3), p.R256C (n=1), p.R258C (n=6), p.G270R (n=2), and p.G368R (n=1). In addition, in comparison to the participants whose medical histories were reviewed, survey participants were slightly older at enrollment (median, 47 versus 44 years; P=0.026) and more likely to have ASCVD events (28.6% versus 9.8%; P<0.001), while aortic events were more frequent in the survey group, but this difference was not statistically significant (50.8% versus 38.0%; P=0.098; Table S5).

In the overall cohort of *ACTA2* cases, ASCVD demonstrated a statistically significant positive correlation with both hyperlipidemia (ρ=0.2489; P=0.0031) and hypertension (ρ=0.2810; P=0.0006; Table S6). In contrast, ASCVD did not exhibit a significant association with diabetes (ρ=−0.0786; P=0.3597), smoking (ρ=0.0949; P=0.2480), or aortic events (ρ=0.0123; P=0.8730). When restricting the analysis to *ACTA2* variants that have at least 1 individual with ASCVD, the correlations

with hyperlipidemia (ρ=0.2674; *P*=0.0108) and hypertension (ρ=0.2550; *P*=0.0122) remained significant.

Assay of *ACTA2* PVs and Correlation With Early Onset ASCVD

We previously showed that WT *Acta2* stably expressed in immortalized mouse *Acta2*$^{-/-}$ SMCs is transcribed, SMA monomer levels are present, and HSF1 is not activated.[14] In contrast, when *Acta2* p.R149C is expressed in a similar manner, the gene is transcribed, but little SMA monomer is detected, and HSF1 is activated.[14] Furthermore, the expression of *ACTA2* p.R149C in *Acta2*$^{-/-}$ SMCs triggers HSF1 activation that subsequently drives cholesterol biosynthesis, in part through augmented HMGCR activation, leading to elevated cellular CE levels, PERK activation, and augmented SMC phenotypic modulation. To identify *ACTA2* missense variants that similarly activate HSF1 and these downstream pathways, the *ACTA2* missense variants occurring in ≥5 cases, plus p.R149H, were individually introduced into mouse *Acta2,* cloned into a lentivirus expression vector, and stably expressed in mouse *Acta2*$^{-/-}$ SMCs (Figure 2A and 2B). All 14 of the *ACTA2* constructs were translated at levels similar to WT *Acta2* except p.R212Q, which had undetectable levels of SMA on immunoblot analyses, and p.R39G, p.R39H, p.R149C, p.R118Q, and p.R258H, which had minimal levels of SMA detected (note that p.R39H also migrated at a higher molecular weight; Figure 2A and 2B; Figure S2A). For p.R149C, along with p.R39H, p.R149H, p.R212Q, and p.R258H, HSF1 expression and transcriptional activation levels were significantly increased (Figure 2C; Figure S2B). All mutant SMAs that activated HSF1 also exhibited significant increases in HMGCR expression, enzymatic activity, and cellular CE levels, along with evidence of PERK activation based on increased *Atf4* expression and KLF4 transcriptional activation (Figure 2D through 2F; Figure S2C and S2D). Interestingly, p.R39G and p.R118Q missense variants had decreased SMA levels but did not activate HSF1; in fact, p.R118Q had significantly decreased activation of HSF1 compared with WT SMCs. The missense variants that activated HSF1 were the only variants that increased the expression of SMC phenotypic modulation markers based on increased expression of *Lgals3, Spp1*, and *Fn1* (Figure 2G; Figure S2E and S2F). There were no differences in the expression of contractile markers *Tagln* and *Cnn1* in any of the variants, regardless of their association with early onset ASCVD (Figure 2G and 2H). It is important to note that decreased CE levels were identified in SMCs expressing p.R39C, p.R39G, p.R118Q, p.G160D, and p.R258C compared with WT SMA levels; all variants were not seen with ASCVD. Finally, *ACTA2* variants p.E59K, p.Y135H, p.P166T, p.R198C, and p.I371N were not present in ≥5 cases, but early onset ASCVD was identified in a carrier, so these variants were also expressed in *Acta2*$^{-/-}$ cells. These variants were expressed at comparable levels as WT *Acta2*, but the protein levels for all 5 mutant SMAs were undetectable on immunoblot analyses (Figure 3A and 3B; Figure S3A). Similar to all other variants seen with early onset ASCVD, these 5 variants also exhibited significantly elevated HSF1 mRNA expression and transcriptional activation, increased expression and enzymatic activity of HMGCR and cellular CE levels, upregulation of *Atf4* expression and KLF4 transcriptional activation, and significant elevation in the mRNA levels of atherosclerosis-associated phenotypic modulation markers (Figure 3C through 3G; Figure S3B through S3H).

Among the 146 patients carrying *ACTA2* variants seen with ASCVD, the prevalence of early onset ASCVD was 23.3%. Correlation analysis between SMA levels and presence of early onset ASCVD for all variants with ≥5 cases was not significant (*P*=0.1014, the Mann-Whitney *U* test). In contrast, significant correlations were obtained between early onset ASCVD and the following: HSF1 activation (*P*=0.035, the Mann-Whitney *U* test), increased HMGCR enzymatic activity (*P*=0.0017, the Student *t* test followed by the Welch correction), increased SMC CE levels (*P*=0.0031, the Student *t* test followed by the Welch correction), and increased SMC phenotypic modulation (assessed by expression of modulation markers *Lgals3*, *Spp1*, and *Fn1*; *P*=0.0117, the Mann-Whitney *U* test). Finally, a significant correlation was identified between the presence of one family member with early onset ASCVD and HSF1 activation (ρ=0.410; *P*=0.0001).

Finally, we sought to determine if explanted dermal fibroblasts from multiple patients with *ACTA2* variants seen with ASCVD (n=1 for p.R118Q and p.R258C, n=2 for p.R185Q, p.R212Q, and p.R258H, and n=4 for p.R149C, along with 3 healthy controls) showed similar activation of HSF1 and the downstream signaling pathways to determine if explanted fibroblasts from patients could also be used to assess for a risk for ASCVD. *ACTA2* expression was higher than that of control fibroblasts in p.R118Q (1 of 1 sample), p.R149C (2/4), p.R185Q (1/2), and p.R258H (1/2). Overall, fibroblasts from individuals with *ACTA2* variants seen with ASCVD exhibited increased *HSF1* expression and transcriptional activation, increased expression and enzymatic activity of HMGCR, elevated intracellular CE levels, upregulated *ATF4* expression, increased *KLF4* expression and transcriptional activation, and increased expression of the atherosclerosis-associated SMC modulation markers, *LGALS3, FN1*, and *SPP1*, compared with fibroblasts isolated from 3 healthy individuals (Figure 4A through 4F; Figure S4A through S4H). One notable exception to this pattern was p.R118Q, which exhibited increased HMGCR expression and activity, CE levels, and expression of phenotypic modulation markers (except *FN*1) despite not upregulating HSF1 and KLF4 expression or transcriptional activity. Expression of *ACTA2* was higher

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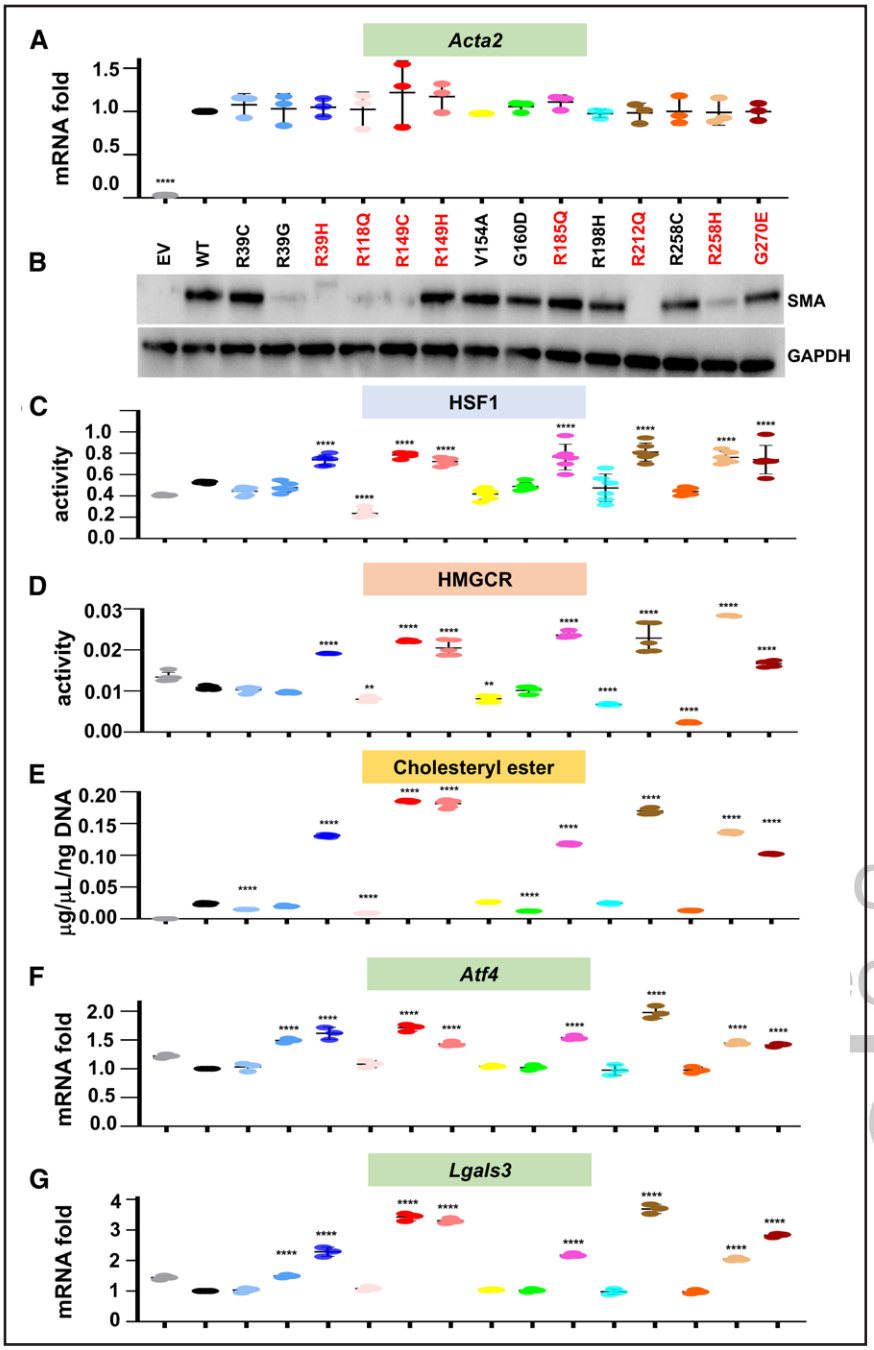

Figure 2. Recurrent *ACTA2* pathogenic/likely pathogenic variants (PVs) seen with early onset atherosclerotic cardiovascular disease (ASCVD) activate HSF1 (heat shock factor 1), increase cholesteryl ester (CE) levels, and augment smooth muscle cell (SMC) phenotypic modulation.

All *Acta2* variants with ≥5 carriers were expressed in *Acta2*$^{-/-}$ SMCs. For all constructs, comparable levels of *Acta2* transcript were expressed (**A**), but some variants exhibited reduced or undetectable levels of SMA (α-smooth muscle actin) monomer on immunoblot analyses (**B**). HSF1 activation (**C**), enzymatic activity of HMGCR (3-hydroxy-3-methylglutaryl-coenzyme A reductase; **D**), and cellular CE levels (**E**) were assayed for the *Acta2* variants, along with expression of *Atf4* (**F**) and the SMC modulation marker, *Lgals3* (**G**). All data presented here are representative of 3 independent experiments. All cellular data passed normality and were analyzed by 1-way ANOVA followed by the Dunnett multiple comparisons test and compared with wild-type (WT) *Acta2*. ** and ****: $P<0.01$ and $P<0.0001$, respectively. EV indicates empty vector.

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in some variants compared with controls, while those of *TAGLN* and *CNN1* were upregulated in some variants and downregulated in others.

DISCUSSION

The molecular mechanism linking a missense PV in *ACTA2*, encoding an SMC contractile protein, to early onset ASCVD in the absence of risk factors, provided the basis for an assay to determine which *ACTA2* PVs are associated with early onset ASCVD.[14] Here, clinical data on ASCVD and results of this mechanism-based assay confirmed that *ACTA2* missense variants that activated HSF1 and accumulated CEs in SMCs are highly correlated with early onset ASCVD in ≥1 individual harboring the corresponding variants. In this assay, the most significant correlation with early onset ASCVD in the cellular assays was with CE levels in SMCs. In addition, these data provide confirmation that the molecular pathway downstream of HSF1 activation is responsible for the early onset ASCVD associated with this gene.[12–14]

Equally as important, this assay delineates which *ACTA2* missense variants trigger early onset ASCVD, information that is critical, given that statin therapy directly targets the molecular pathogenesis of early

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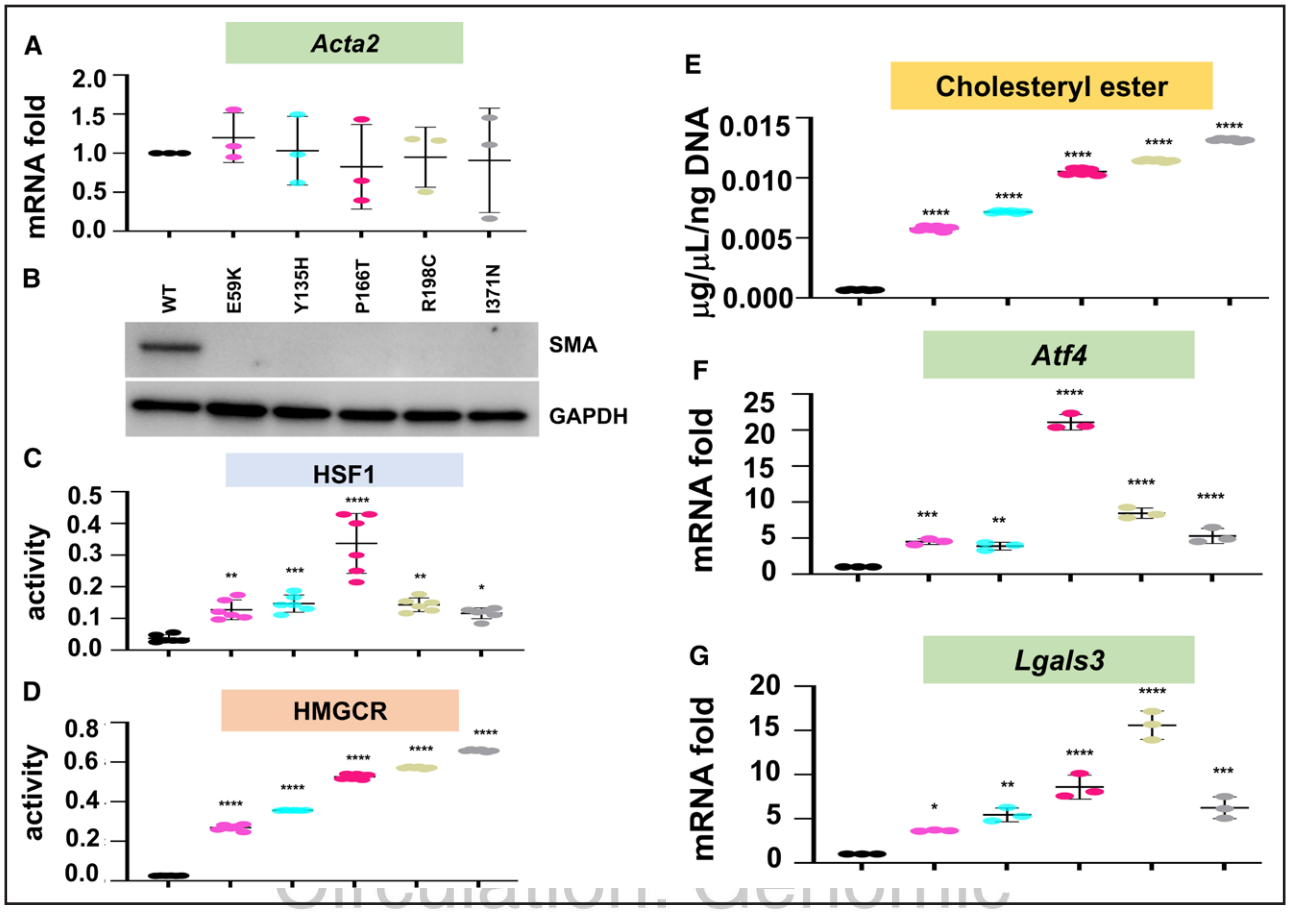

Figure 3. *ACTA2* pathogenic/likely pathogenic variants (PVs) present in <5 cases and seen with early onset atherosclerotic cardiovascular disease (ASCVD) also activate HSF1 (heat shock factor 1), increase cholesteryl ester (CE) levels, and augment smooth muscle cell (SMC) phenotypic modulation.

A and **B**, *Acta2* variants with <5 patients and seen with early onset ASCVD were expressed in *Acta2*$^{-/-}$ SMCs and show comparable expression of *Acta2* mRNA and undetectable levels of SMA (α-smooth muscle actin) protein compared with wild-type (WT) *Acta2*. All 5 *Acta2* missense variants that predispose to ASCVD upregulate HSF1 activation (**C**), increase the enzymatic activity of HMGCR (3-hydroxy-3-methylglutaryl-coenzyme A reductase; **D**), elevate the levels of cellular CEs (**E**), and induce expression of *Atf4* (**F**) and the SMC modulation marker *Lgals3* (**G**). All data presented here are representative of 3 independent experiments. All cellular data passed normality and were analyzed by 1-way ANOVA followed by the Dunnett multiple comparisons test and compared with WT *Acta2*. *, **, ***, and ****: $P<0.05$, $P<0.01$, $P<0.001$, and $P<0.0001$, respectively.

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onset ASCVD, even in individuals with normal blood/serum cholesterol levels. All *ACTA2* missense mutants that activated HSF1 also increased HMGCR activity and cellular CE levels. High-fat diet–fed *Acta2*$^{R149C/+}$*Apoe*$^{-/-}$ mice were treated with pravastatin to prevent the augmented cholesterol synthesis in SMCs, and this treatment decreased plaque burden in these mutant mice to levels similar to the *Apoe*$^{-/-}$ mice without differences in plasma lipid levels between these mice.[14] The fact that ASCVD has been observed in *ACTA2* p.R149C patients in their 20s, and individuals have had CAD events in their early 30s, leads to the recommendation that statin therapy be considered by the age of ≤20 years in *ACTA2* patients harboring variants that increase CE levels. While we used pravastatin in our previous mouse studies, any statin should be effective in humans because all statins work by targeting HMGCR. In patients with the *ACTA2* p.R149C variant, early onset CAD is the most common presentation, but the ASCVD can involve other arteries and the aorta. It is also notable that 9 of 47 *ACTA2* p.R149C patients had both an aortic event and early onset atherosclerosis, illustrating that it is not uncommon for a patient to have both vascular diseases.

Significant correlations were identified between ASCVD in patients with *ACTA2* PVs and hypertension and hyperlipidemia, but not with diabetes or smoking. Hypertension is known to activate HSF1 in the aortic wall and can potentially further augment cholesterol biosynthesis, PERK activation, and SMC phenotypic modulation.[28] Similarly, high circulating blood cholesterol levels could contribute to SMC intracellular CE levels and augment PERK activation and downstream pathways. Thus, these risk factors for ASCVD could augment the risk for *ACTA2* PV carriers. Importantly, any stimulus or genetic variant that induces cytosolic stress is expected to activate the same pathway, as illustrated by studies that

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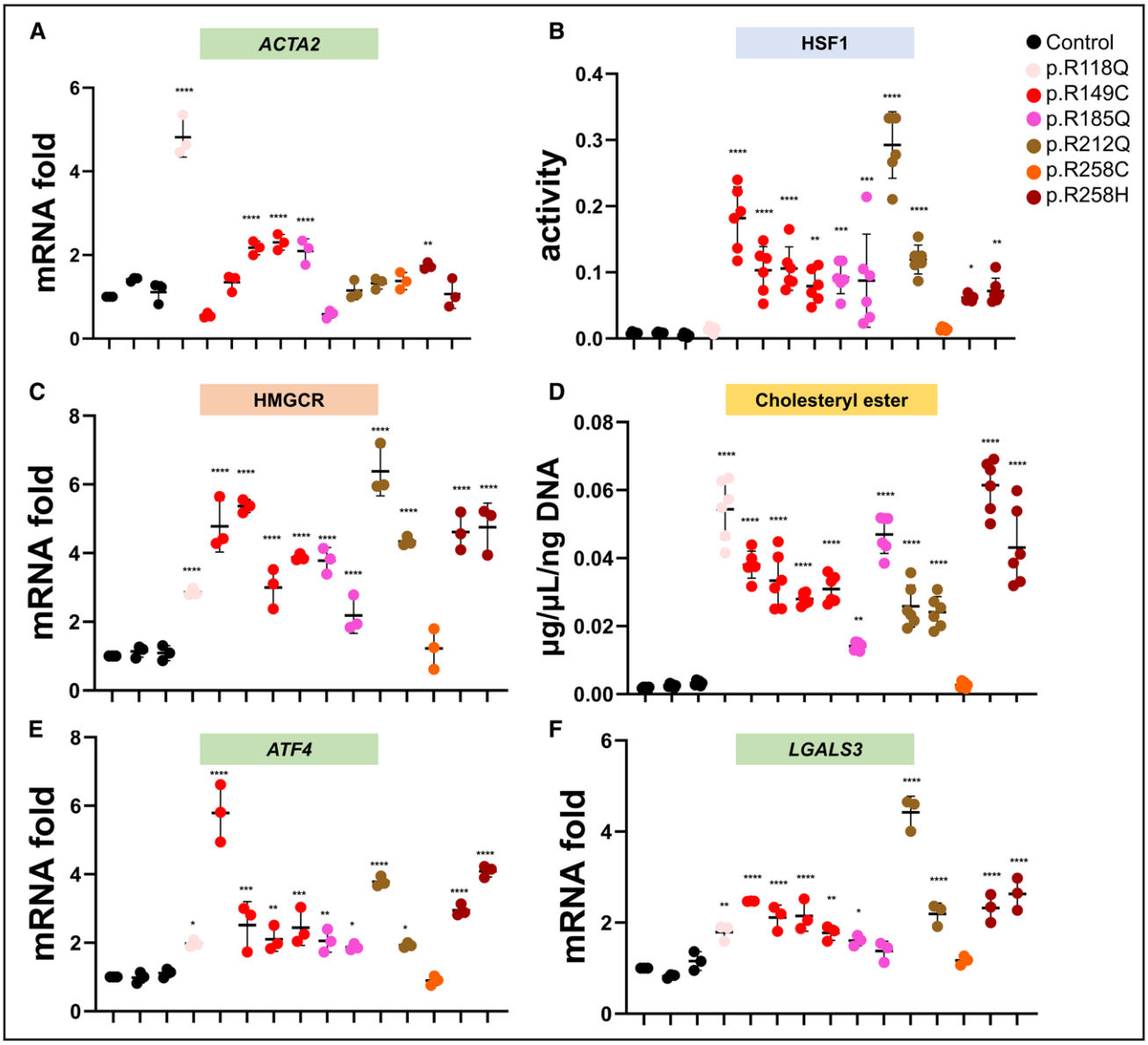

Figure 4. Dermal fibroblasts from individuals with *ACTA2* pathogenic variants seen with atherosclerotic cardiovascular disease (ASCVD) upregulate HSF1 (heat shock factor 1) signaling, cholesterol biosynthesis, and smooth muscle cell (SMC) phenotypic modulation.

A, Fibroblasts isolated from individuals harboring *ACTA2* pathogenic variants show variable *ACTA2* mRNA expression compared with healthy individuals. Fibroblasts from individuals with *ACTA2* pathogenic variants that have been seen with ASCVD upregulate HSF1 activity (**B**), *HMGCR* activity (**C**), intracellular cholesteryl ester (CE) levels (**D**), *ATF4* expression (**E**), and expression of the SMC modulation marker *LGALS3* (**F**). All data passed normality and were analyzed by 1-way ANOVA followed by the Dunnett multiple comparisons test and compared with the first bar among the control samples. All variants that have been seen with ASCVD are designated in red in the legend (next to **B**). *, **, ***, and ****: $P<0.05$, $P<0.01$, $P<0.001$, and $P<0.0001$, respectively.

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heat shock control SMCs can activate HSF1, HMGCR, and cholesterol biosynthesis, as well as downstream PERK signaling and SMC phenotypic modulation.[14] It is also important to note that PVs in other genes can also activate this pathway in SMCs and lead to early onset ASCVD.[29] Taken together, while our data establish *ACTA2* missense variants activating HSF1 and cholesterol biosynthesis as a risk for early onset ASCVD, contributions from elevated plasma lipid levels in increasing ASCVD risk cannot be ruled out. The effect of statins on vascular SMCs might have a direct role in mitigating ASCVD risk, but their effects on plasma lipid levels might also contribute to the overall beneficial effects against ASCVD. Future studies will focus on identifying ideal therapeutic strategies for these patients.

In an unstressed cell, HSF1 remains sequestered by the CCT complex, and inactivation or dysfunction of the CCT complex can release and activate HSF1.[30] We previously determined that SMA with the p.R149C variant is misfolded and retained in the CCT complex, leading

to activation of HSF1.[14,15] All actins are folded into their native conformation by the CCT complex, with specific amino acid residues being responsible for the different steps of the process that involves substrate recognition, transition to correct folding intermediates, and release of the native actin from the CCT complex.[16,31] While no direct studies have been performed using SMA, kinetic studies of CCT folding have been done after mutagenesis of specific amino acid residues of β-actin, which has ≈95% sequence homology with SMA. These studies identified residues neighboring amino acids 125, 215, 255, 260, 340, and 360 as being crucial for correct folding and successful release of the folded protein from the CCT complex.[31] Multiple *ACTA2* variants tested in this study are adjacent to residues critical for the correct folding of β-actin and potentially lead to retention in the CCT complex in a similar manner.

In our cellular data using *Acta2*$^{-/-}$ SMCs, we observed robust upregulation of SMC phenotypic modulation markers, but these cells did not exhibit decreased levels of contractile genes (*Tagln* and *Cnn1*) to the same extent as in *Acta2*$^{R149/+}$ SMCs.[14] We previously observed that *Acta2*$^{-/-}$ SMCs have increased expression of contractile proteins at baseline, whereas the *Acta2*$^{R149/+}$ SMCs do not.[32] The fact that the *Acta2*$^{-/-}$ SMCs have activated pathways that increase the expression of contractile proteins and the *Acta2*$^{R149/+}$ SMCs do not is most likely the cause of the lack of dedifferentiation of the *Acta2*$^{-/-}$ SMCs when HSF1 is activated. The reason for this variability in contractile protein levels is unknown but may be due to the fact that fibroblasts exposed to TGFβ in culture transition to myofibroblasts when they are proliferating, during which the cells increase expression of SMC markers.[33] Therefore, the fibroblast cultures may be variable in terms of their contractile markers based on how quiescent they are when the assay is performed.

Different missense variants can alter the same amino acid in a protein to multiple other amino acids. An unexpected observation in this study was the fact that the exact amino acid change resulting from the variant plays a role in whether the variant is associated with early onset ASCVD. For example, *ACTA2* p.R39 can be altered to a cysteine, histidine, or glycine, but only p.R39C expression was associated with the production of normal levels of SMA. In contrast, SMA levels were decreased for both p.R39G and p.R39H, but HSF1 was only activated with the p.R39H variant, and early onset ASCVD is only observed in cases with the p.R39H variant. Mutagenesis of β-actin identified that a region around amino acid 30 caused the protein to be misfolded but still released from the CCT complex.[31] Thus, we hypothesize that p.R39C is not misfolded, p.R39G is misfolded and degraded but fails to activate HSF1, and p.R39H is similarly degraded but does activate HSF1. For the 2 *ACTA2* p.R258 variants, p.R258H expression leads to lower SMA levels, HSF1 activation, and predisposition to early onset ASCVD, whereas p.R258C is associated with normal levels of SMA and no HSF1 activation and is not associated with early onset atherosclerosis. Although many of the variants expressed were associated with decreased SMA levels in the assay, p.R185Q and p.G270E were not, but they still activated HSF1 signaling and predisposed to ASCVD. These results aligned with our correlation studies, in which assessing levels of SMA in our *Acta2*$^{-/-}$ SMC assay system of the individual missense variants was not as highly associated with early onset ASCVD as were HSF1 activation and CE levels. Conversely, the p.R118Q variant was the only variant that exhibited reduced SMA levels but did not activate HSF1 or downstream signaling. As this variant is seen with ASCVD, perhaps the atherosclerosis is driven by another mechanism or HSF1 is activated in SMCs in vivo.[20] We postulate that in *ACTA2* PVs that do not activate HSF1 but are associated with decreased SMA levels, the mutant protein is folded correctly so that HSF1 is not activated, but mutant SMA is degraded at a later stage at a faster rate than WT SMA. Finally, in the assay, the p.R149H variant activated HSF1 despite the production of robust SMA levels, and we had previously shown that SMCs explanted from the aorta of an *ACTA2* p.R149H patient activated HSF1 and the downstream pathways.[14] Because only 1 individual has been identified with this variant, identification of additional individuals with this variant will test its association with early onset ASCVD.

We previously determined that HSF1 activation of downstream signaling pathways underlies augmented atherosclerosis in a mouse model with SMC-specific deficiency of the centrosomal protein, pericentrin, and the increased plaque burden can also be successfully reversed by treating the mice with a statin.[29] Fibroblasts explanted from 12 patients with *ACTA2* PVs demonstrated HSF1 and HMGCR activation, CE accumulation, PERK signaling, and increased expression of atherosclerosis-associated phenotypic modulation markers in the variants seen with ASCVD compared with 3 healthy individuals and 1 patient with a variant not seen with ASCVD. These observations indicate that dermal fibroblasts from patients who have early onset ASCVD and normal serum cholesterol levels could potentially be used to assay HSF1 activation and cellular CE accumulation. In other words, fibroblasts could be used to determine if an underlying, unidentified genetic variant is activating this pathway and to provide justification for statin therapies in such individuals despite normal serum cholesterol levels. Interestingly, while all fibroblasts with variants seen with ASCVD, except p.R118Q demonstrated an overall pattern of increased HSF1 signaling, cholesterol accumulation, and PERK signaling leading to the upregulation of phenotypic modulation markers, the expressions of contractile markers such as *ACTA2*, *TAGLN*, and *CNN1* were highly variable. The p.R118Q fibroblasts

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upregulated HMGCR, cholesterol levels, *ATF4* expression, and some markers of SMC phenotypic modulation, without increases to HSF1 and KLF4 expression or activity, while lentiviral overexpression of the p.R118Q variant in *Acta2*$^{-/-}$ SMCs did not induce HSF1 activation or downstream signaling. These data suggest that ASCVD associated with this PV may be driven by different pathways from HSF1 signaling. Because we had variable results with the *ACTA2* p.R118Q in different systems, obtaining SMCs heterozygous for this variant would be the first step towards understanding how this variant triggers ASCVD.

These analyses were limited by small patient cohort sizes, and we could not access all the medical records, which could have corroborated a greater number of ASCVD diagnoses. Despite these limitations, our molecular data clearly associate specific *ACTA2* variants with upregulated HSF1 activity and cellular CE levels, which helps to address these limitations. Because increased CE levels trigger SMC phenotypic modulation to chondrocyte-like cells, cardiac calcium scores may potentially be useful for confirmation of early onset ASCVD in individuals with *ACTA2* variants and inform us as to when statin therapy should be started. However, the coronary plaques of *ACTA2* PV patients may not contain significant calcium deposits, and further studies are needed to determine whether coronary calcium scores correlate with ASCVD risk in patients with *ACTA2* PVs.

In summary, we have delineated the specific *ACTA2* variants triggering early onset ASCVD through clinical data and a functional assay based on the molecular pathogenesis responsible for the ASCVD. Statin therapy should be considered for all patients with *ACTA2* PVs that increase the risk for ASCVD, even in the absence of hypercholesterolemia. Importantly, surveillance and timely clinical treatment can prevent both acute type A aortic dissections and ASCVD complications in individuals with *ACTA2* PVs. Thus, these data illustrate the use of molecular assays to inform personalized medical care for individuals with *ACTA2* PVs, with the ultimate goal of improving outcomes for these individuals.

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Received February 20, 2025; accepted January 8, 2026.

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Acknowledgments

The authors would like to thank the patients who participated in this study.

Sources of Funding

This work was supported by the National Heart, Lung, and Blood Institute (grant R01 HL146583), an American Heart Association Merit Award, funding from the Genetic Aortic Disorders Association Canada and the John Ritter Foundation (to Dr Milewicz), and an American Heart Association Career Development Award (grant 24CDA1273174 to Dr Chattopadhyay).

Disclosures

None.

Supplemental Material

Tables S1–S6
Figures S1–S4
Excel Files S1–S2
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